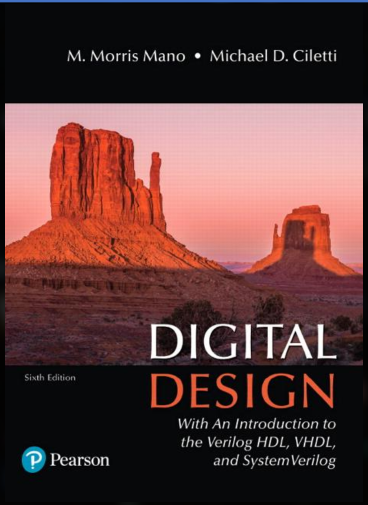
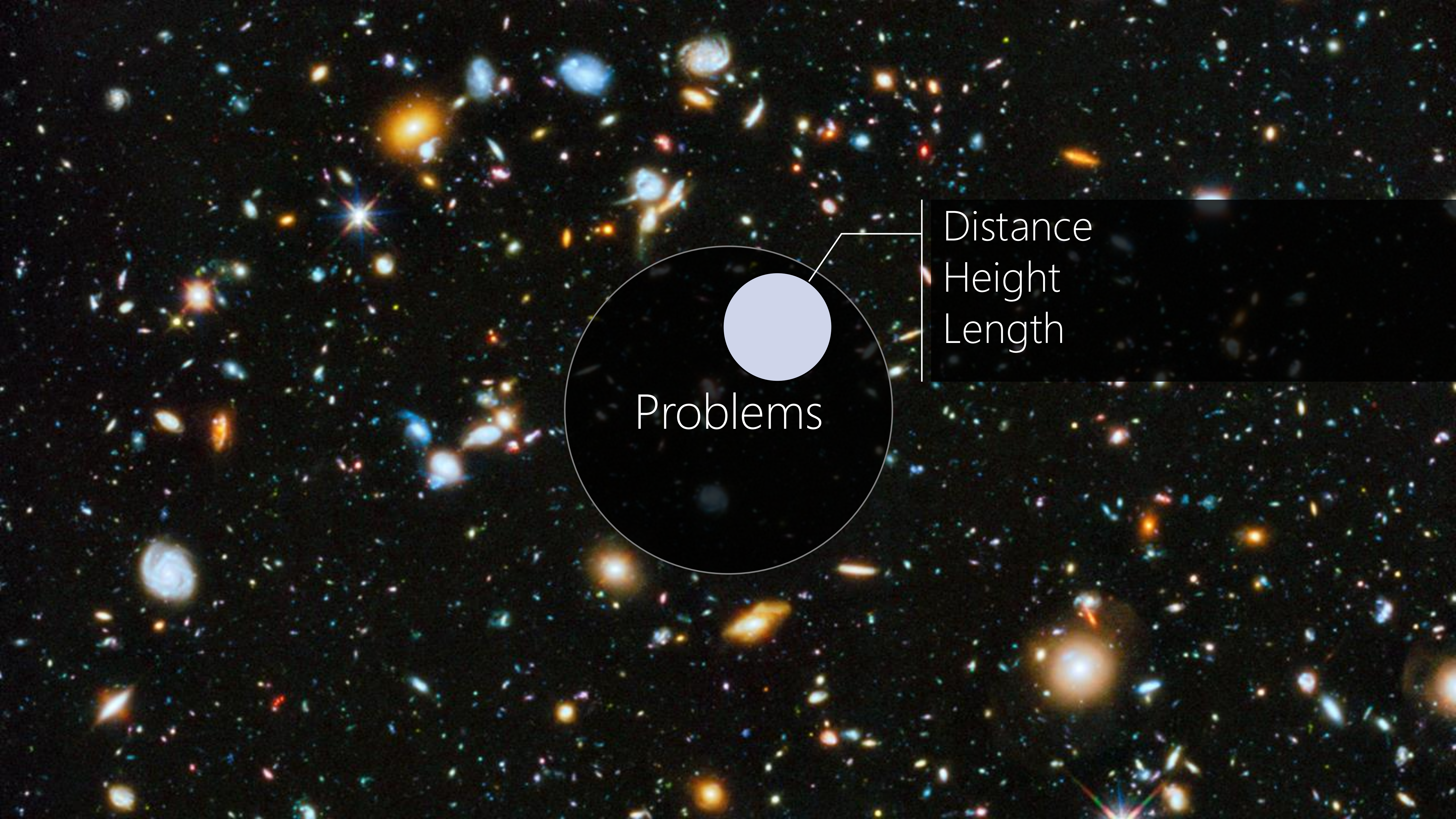
LEC03 & LAB03

Week03 >> Lec03: Signed Numbers
Week03 >> Lab03: Signed Complement



Chapter 1
Digital Systems and Binary Numbers


Problems
Distance
Height
Length

Quantization

NUMBER SYSTEMS

QUATERNARY SYSTEM

/kwaa·**tur**·neh·ree/

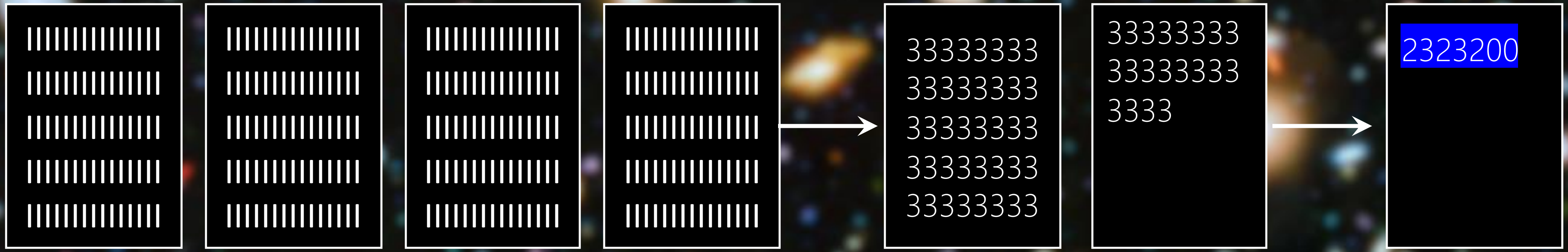
aka. Base-4, Radix-4

$(0,1,2,3)_4$

Hindu-Arabic Numerals
Originated in India
7th Century AD

$$N = 12{,}000 \rightarrow (2323200)_4$$

We'll see how to convert from decimal to base-4 or any other number systems later. Stay tuned!



COMMON NUMBER SYSTEMS

BINARY | BASE-2 | RADIX-2

$(0,1)_2$

OCTAL | BASE-8 | RADIX-8

$(0,1,2,3,4,5,6,7)_8$

DECIMAL | BASE-10 | RADIX-10

$(0,1,2,3,4,5,6,7,8,9)_{10}$

Arabic 0123456789

Eastern Arabic ٠١٢٣٤٥٦٧٨٩

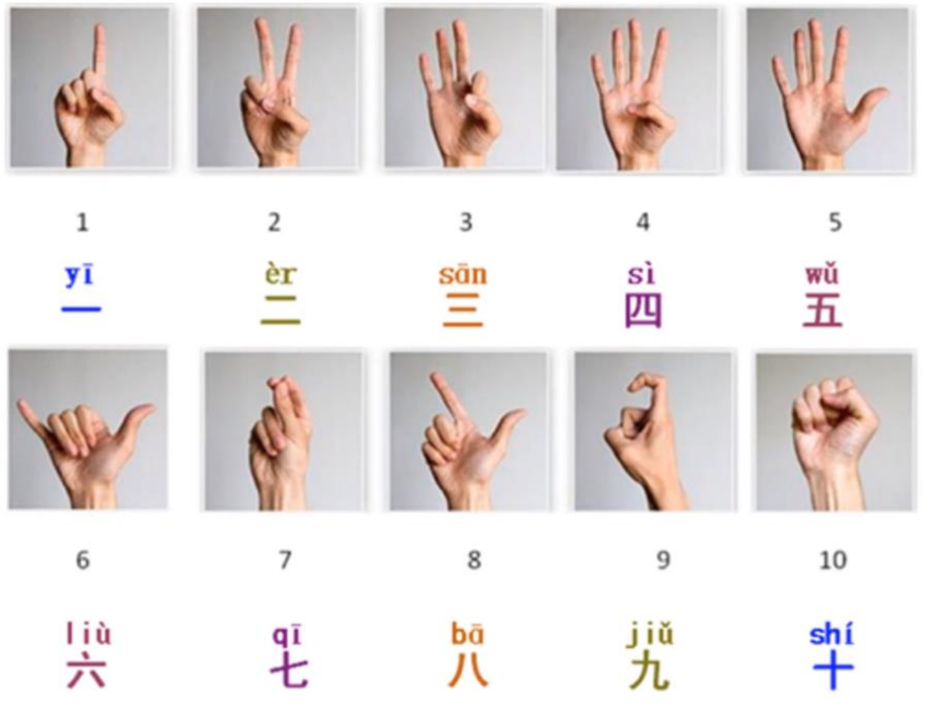
"the *hand makes the two complementary aspects of integers entirely intuitive*. It serves as an instrument permitting natural movement between cardinal and ordinal numbering. If you need to show that a set contains three, four, seven or ten elements, you raise or bend *simultaneously* three, four, seven or ten fingers, using your hand as cardinal mapping. If you want to count out the same things, then you bend or raise three, four, seven or ten fingers in *succession*, using the hand as an ordinal counting tool."

- Georges Ifrah The Universal History of Numbers (Wiley, 2000, pp. 21-22)

Thanks Cecelia Nydam and Giavi Tran for the hint!

Thai ๐๑๒

Chinese 〇一二三



Weird NUMBER SYSTEMS

HEXADECIMAL | BASE-16 | RADIX-16

$(0,1,2,3,4,5,6,7,9,A,B,C,D,E,F)_{16}$

BASE-64 | RADIX-64

$(A,B,C, \ldots,Z, a,b,c,\ldots,z,0,1,2,\ldots,9,+,/)_{64}$

1992: RFC 1341

MIME (MULTIPURPOSE INTERNET MAIL EXTENSIONS)

Mechanisms For Specifying And Describing The Format Of Internet Message Bodies

Digit	Value
A	0
B	1
C	2
D	3
E	4
F	5
G	6
H	7
I	8
J	9
K	10
L	11
M	12
N	13
O	14
P	15

→

Digit	Value
Q	16
R	17
S	18
T	19
U	20
V	21
W	22
X	23
Y	24
Z	25
a	26
b	27
c	28
d	29
e	30
f	31

→

Digit	Value
g	32
h	33
i	34
j	35
k	36
l	37
m	38
n	39
o	40
p	41
q	42
r	43
s	44
t	45
u	46
v	47

→

Digit	Value
w	48
x	49
y	50
z	51
0	52
1	53
2	54
3	55
4	56
5	57
6	58
7	59
8	60
9	61
+	62
/	63

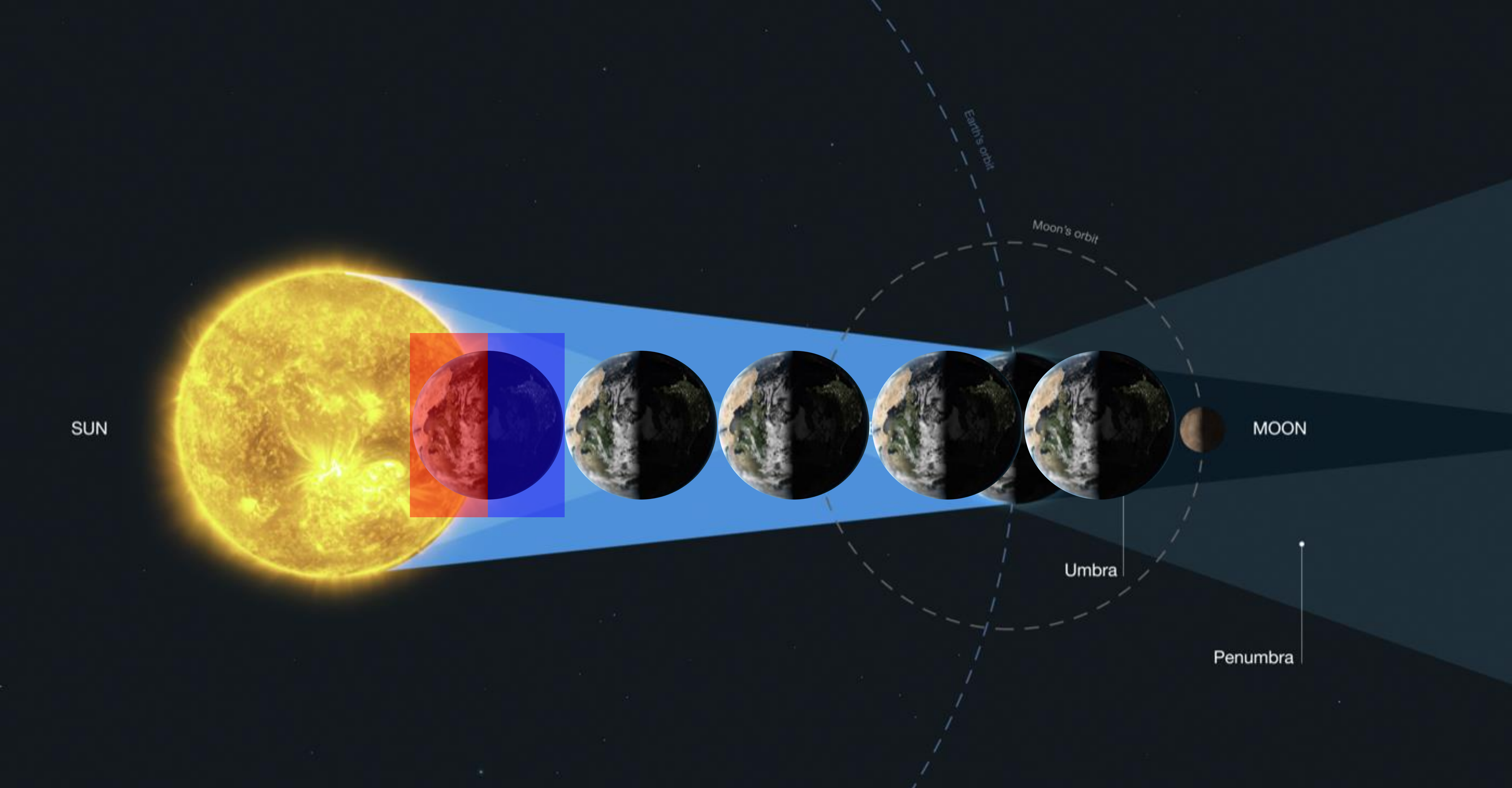
64^7	64^6	64^5	64^4	64^3	64^2	64^1	64^0	
3	a	A	/	d	1	H	+	$\times$
55 $\times 64^7$	26 $\times 64^6$	0 $\times 64^5$	63 $\times 64^4$	29 $\times 64^3$	53 $\times 64^2$	7 $\times 64^1$	62 $\times 64^0$	$\sum$
								243,680,329, 290,238

RADIX-R NUMBER SYSTEM

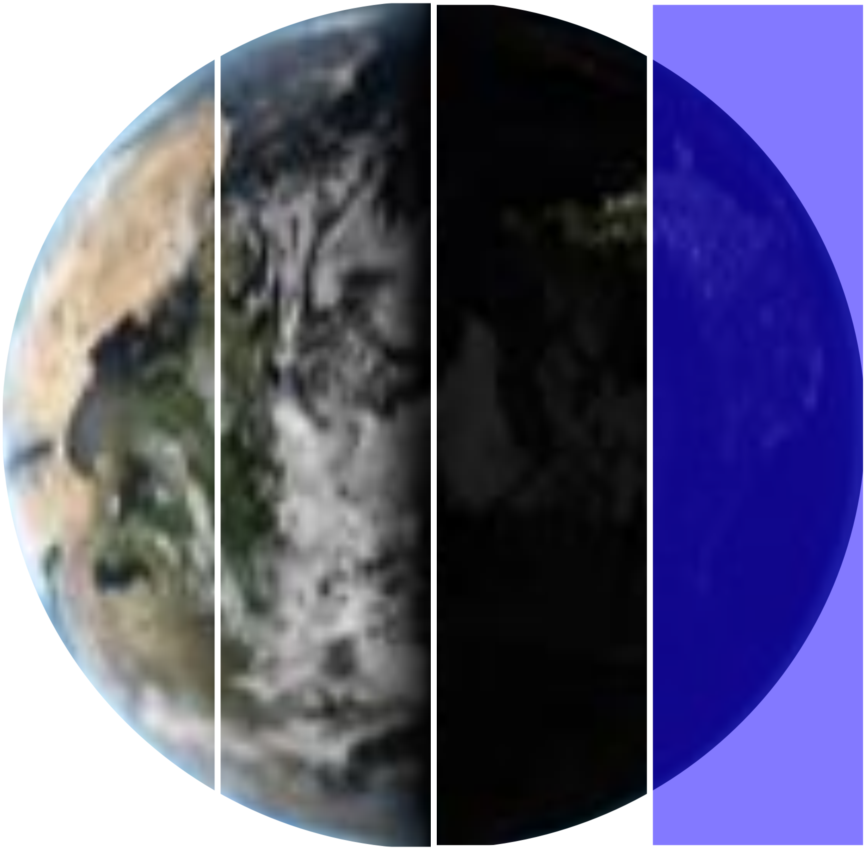
aka. Base-r Number System

Hossein's number system is not a Radix-r number system!

FRACTION

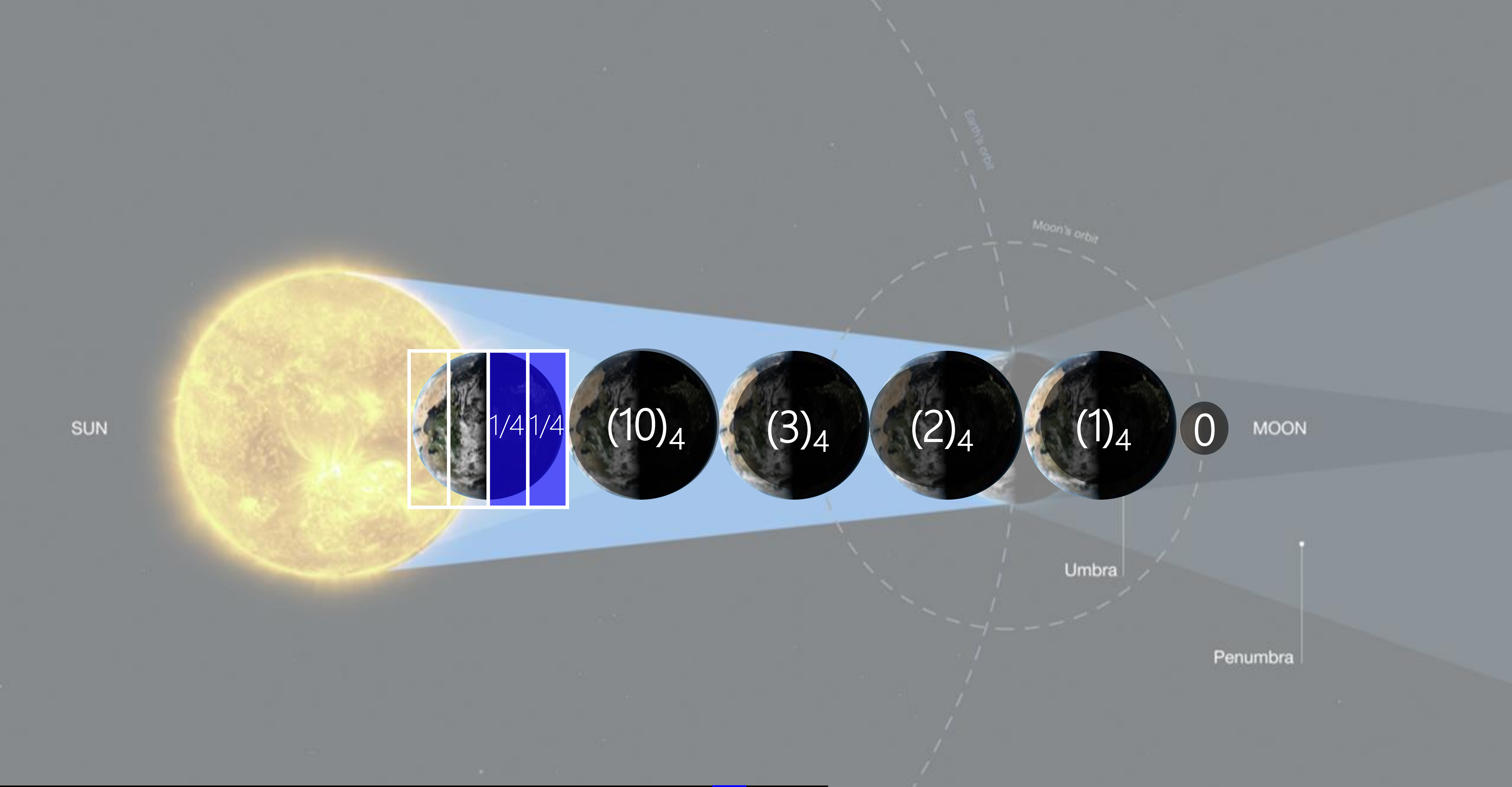


Moon's Distance to Sun: $(10.?)_4 = (4.5)_{10}$

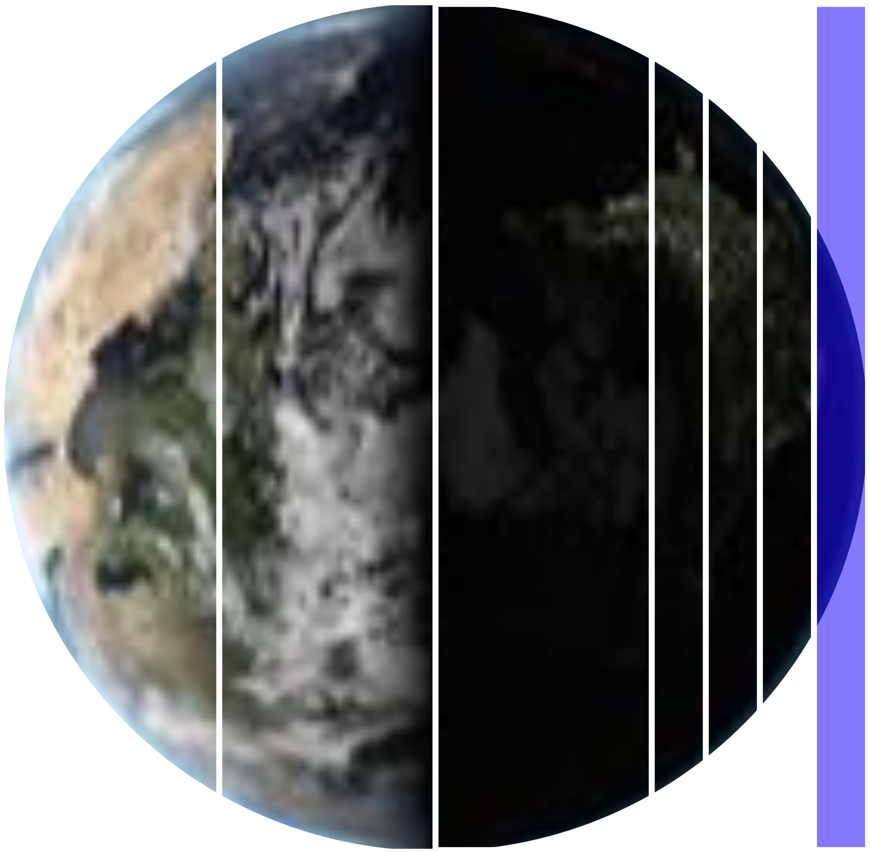


Fraction Point

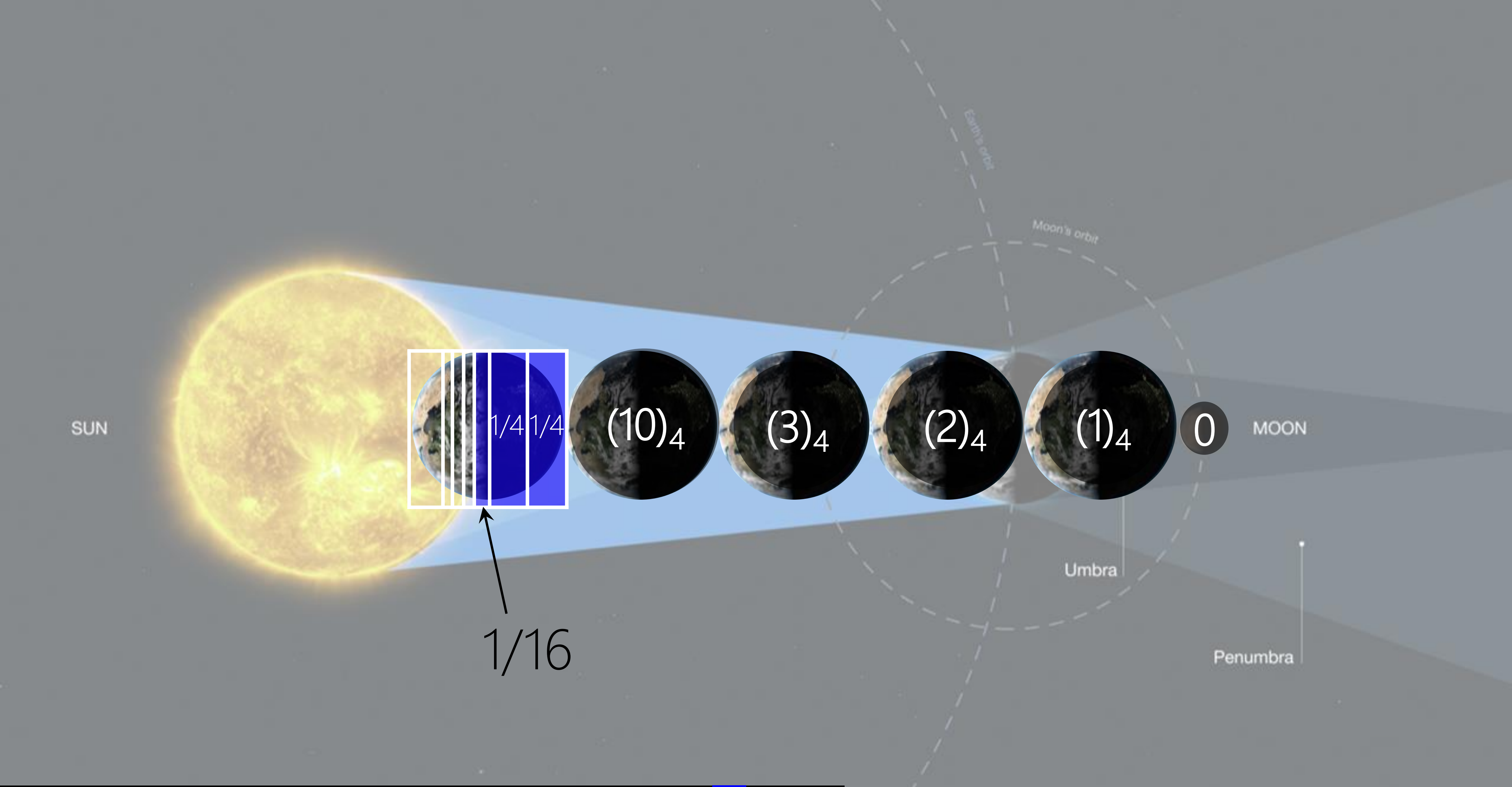
Radix-4 (Base-4) = 1/4 Earth = 4^{-1} Earth = $(.1)_4$



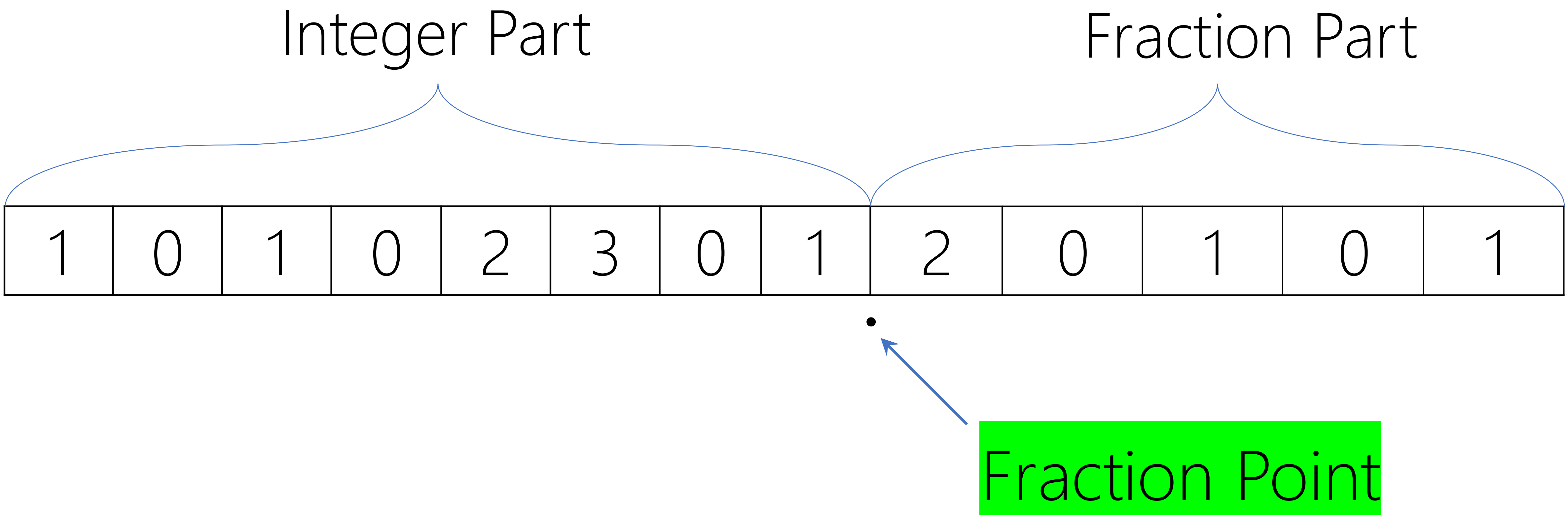
Moon's Distance to Sun in Radix-4: $(10.2)_4$



Radix-4 (Base-4) = (1/4)/4 Earth = 1/16 Earth = 4^{-2} Earth



Moon’s Distance to Sun in Radix-4: $(10.21)_4$



We don’t waste a position for the fraction point.
It’s location is defined already.

Integer Part

4^7	4^6	4^5	4^4	4^3	4^2	4^1	4^0
1	0	1	0	2	3	0	1
1 $\times$16,384	0	1 $\times$1,024	0	2 $\times$64	3 $\times$16	0	1
				17,584			

$\times$. $\sum$.

Fraction Part

4^{-1}	4^{-2}	4^{-3}	4^{-4}	4^{-5}
2	0	1	0	1
$\frac{2}{4}$	0	$\frac{1}{64}$	0	$\frac{1}{1{,}024}$
5166015625				

Let $(N)_r$ be a radix-r (base-r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1}\, r^{n-1} + \cdots + d_0 r^0 \,.\, d_{-1}\, r^{-1} + d_{-2}\, r^{-2} + \cdots + d_{-m}\, r^{-m})_{10}$$

Fraction Point

where:

r	=	radix (base)
d_i	=	digit at position i, $0 \le d_i \le r - 1$
r^i	=	weight of position i
n	=	number of digits in integer part of N
m	=	number of digits in fraction part of N

Let $(N)_r$ be a radix-r (base-r) number in a positional weighting number system, then

$$\text{Min} = (0_{n-1} \cdots 0_1 0_0 \,.\, 0_{-1} 0_{-2} \cdots 0_{-m-1} 0_{-m})_r = (0 \,.\, 0)_{10}$$

$$\text{Max} = ((r-1)_{n-1} \cdots (r-1)_0 \,.\, (r-1)_{-1}(r-1)_{-2} \cdots (r-1)_{-m-1}(r-1)_{-m})_r = (r^n-1 \,.\, ?)_{10}$$

$$\text{Unit} = (0_{n-1} \cdots 0_1 0_0 \,.\, 0_{-1} 0_{-2} \cdots 0_{-m-1} 1_{-m})_r = (r^{-m})_{10}$$

where:

- r = radix (base)
- r^i = weight of position i
- n = number of digits in integer part of N
- m = number of digits in fraction part of N

Lecture Assignment

PRACTICE RADIX-2

Radix-2								
Integer (n=4)					Fraction (m=3)			Radix-10
2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1*2^{-3}$ = 1/8 = 0.125
0	0	0	0	.	0	1	0	$1*2^{-2} + 0*2^{-3}$ = 1/4 = 0.25
0	0	0	0	.	0	1	1	$1*2^{-2} + 1*2^{-3}$ = 1/4 + 1/8 = 3/8 = 0.375
0	0	0	0	.	1	0	0	$1*2^{-1} + 0*2^{-2} + 0*2^{-3}$ = ½ = 0.5
0	0	0	0	.	1	0	1	$1*2^{-1} + 0*2^{-2} + 1*2^{-3}$ = 1/2 + 1/8 = 5/8 = 0.625
0	0	0	0	.	1	1	0	$1*2^{-1} + 1*2^{-2} + 0*2^{-3}$ = 1/2 + 1/4 = 3/4 = 0.75
0	0	0	0	.	1	1	1	$1*2^{-1} + 1*2^{-2} + 1*2^{-3}$ = 1/2 + 1/4 + 1/8 = 0.875
0	0	0	1	.	0	0	0	$1*2^{0} + 0*2^{-1} + 0*2^{-2} + 0*2^{-3}$ = 1
0	0	0	1	.	0	0	1	1.125
0	0	0	1	.	0	1	0	1.25
0	0	0	1	.	0	1	1	1.375
0	0	0	1	.	1	0	0	1.5
0	0	0	1	.	1	0	1	1.625
0	0	0	1	.	1	1	0	1.75
0	0	0	1	.	1	1	1	1.875
0	0	1	0	.	0	0	0	2

What is the max in this system with these spaces?

$(1111.111)_2 = (15.875)_{10}$

Radix-2								
Integer (n=4)					Fraction (m=3)			Radix-10
2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1*2^{-3} = 1/8 = 0.125$
0	0	0	0	.	0	1	0	$1*2^{-2} + 0*2^{-3} = 1/4 = 0.25$
0	0	0	0	.	0	1	1	$1*2^{-2} + 1*2^{-3} = 1/4 + 1/8 = 3/8 = 0.375$
0	0	0	0	.	1	0	0	$1*2^{-1} + 0*2^{-2} + 0*2^{-3} = ½ = 0.5$
0	0	0	0	.	1	0	1	$1*2^{-1} + 0*2^{-2} + 1*2^{-3} = 1/2 + 1/8 = 5/8 = 0.625$
0	0	0	0	.	1	1	0	$1*2^{-1} + 1*2^{-2} + 0*2^{-3} = 1/2 + 1/4 = 3/4 = 0.75$
0	0	0	0	.	1	1	1	$1*2^{-1} + 1*2^{-2} + 1*2^{-3} = 1/2 + 1/4 + 1/8 = 0.875$
0	0	0	1	.	0	0	0	$1*2^{0} + 0*2^{-1} + 0*2^{-2} + 0*2^{-3} = 1$
0	0	0	1	.	0	0	1	1.125
0	0	0	1	.	0	1	0	1.25
0	0	0	1	.	0	1	1	1.375
0	0	0	1	.	1	0	0	1.5
0	0	0	1	.	1	0	1	1.625
0	0	0	1	.	1	1	0	1.75
0	0	0	1	.	1	1	1	1.875
0	0	1	0	.	0	0	0	2

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

A. More precision.

A. More fraction positions.

A. More in m! How much?

Radix-2								
Integer (n=4)					Fraction (m=3)			Radix-10
2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1*2^{-3} = 1/8 = 0.125$
0	0	0	0	.	0	1	0	$1*2^{-2} + 0*2^{-3} = 1/4 = 0.25$
0	0	0	0	.	0	1	1	$1*2^{-2} + 1*2^{-3} = 1/4 + 1/8 = 3/8 = 0.375$
0	0	0	0	.	1	0	0	$1*2^{-1} + 0*2^{-2} + 0*2^{-3} = ½ = 0.5$
0	0	0	0	.	1	0	1	$1*2^{-1} + 0*2^{-2} + 1*2^{-3} = 1/2 + 1/8 = 5/8 = 0.625$
0	0	0	0	.	1	1	0	$1*2^{-1} + 1*2^{-2} + 0*2^{-3} = 1/2 + 1/4 = 3/4 = 0.75$
0	0	0	0	.	1	1	1	$1*2^{-1} + 1*2^{-2} + 1*2^{-3} = 1/2 + 1/4 + 1/8 = 0.875$
0	0	0	1	.	0	0	0	$1*2^0 + 0*2^{-1} + 0*2^{-2} + 0*2^{-3} = 1$
0	0	0	1	.	0	0	1	1.125
0	0	0	1	.	0	1	0	1.25
0	0	0	1	.	0	1	1	1.375
0	0	0	1	.	1	0	0	1.5
0	0	0	1	.	1	0	1	1.625
0	0	0	1	.	1	1	0	1.75
0	0	0	1	.	1	1	1	1.875
0	0	1	0	.	0	0	0	2

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

B. Find the closest number

$(1.000)_2 = (1)_{10} =>$ Error = 0.02

$(1.001)_2 = (1.125)_{10} =>$ Error = 0.105

Radix-2								Radix-10
Integer (n=4)					Fraction (m=3)			
2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1*2^{-3} = 1/8 = 0.125$
0	0	0	0	.	0	1	0	$1*2^{-2} + 0*2^{-3} = 1/4 = 0.25$
0	0	0	0	.	0	1	1	$1*2^{-2} + 1*2^{-3} = 1/4 + 1/8 = 3/8 = 0.375$
0	0	0	0	.	1	0	0	$1*2^{-1} + 0*2^{-2} + 0*2^{-3} = ½ = 0.5$
0	0	0	0	.	1	0	1	$1*2^{-1} + 0*2^{-2} + 1*2^{-3} = 1/2 + 1/8 = 5/8 = 0.625$
0	0	0	0	.	1	1	0	$1*2^{-1} + 1*2^{-2} + 0*2^{-3} = 1/2 + 1/4 = 3/4 = 0.75$
0	0	0	0	.	1	1	1	$1*2^{-1} + 1*2^{-2} + 1*2^{-3} = 1/2 + 1/4 + 1/8 = 0.875$
0	0	0	1	.	0	0	0	$1*2^{0} + 0*2^{-1} + 0*2^{-2} + 0*2^{-3} = 1$
0	0	0	1	.	0	0	1	1.125
0	0	0	1	.	0	1	0	1.25
0	0	0	1	.	0	1	1	1.375
0	0	0	1	.	1	0	0	1.5
0	0	0	1	.	1	0	1	1.625
0	0	0	1	.	1	1	0	1.75
0	0	0	1	.	1	1	1	1.875
0	0	1	0	.	0	0	0	2

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! The numbers in this system increments by 0.125 unit.

Solution?

B. Find the closest number

$(1.000)_2 = (1)_{10} =>$ Error = 0.02

$(1.001)_2 = (1.125)_{10} =>$ Error = 0.105

PRACTICE RADIX-4

Radix-4								
Integer (n=4)					Fraction (m=3)			Radix-10
4^3	4^2	4^1	4^0	.	4^{-1}	4^{-2}	4^{-3}	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1*4^{-3} = 1/64 = 0.015625$
0	0	0	0	.	0	0	2	$2*4^{-3} = 2/64 = 0.03125$
0	0	0	0	.	0	0	3	$3*4^{-3} = 3/64 = 0.046875$
0	0	0	0	.	0	1	0	$1*4^{-2} + 0*4^{-2} = 1/16 = 0.0625$
0	0	0	0	.	0	1	1	$1*4^{-2} + 1*4^{-2} = 1/16 + 1/64 = 0.078125$
0	0	0	0	.	0	1	2	$1*4^{-2} + 2*4^{-2} = 1/16 + 2/64 = 0.09375$
0	0	0	0	.	0	1	3	$1*4^{-2} + 3*4^{-2} = 1/16 + 3/64 = 0.109375$
0	0	0	0	.	0	2	0	$2*4^{-2} + 0*4^{-2} = 2/16 = 0.125$
...								
0	0	0	0	.	3	3	3	$3*4^{-1} + 3*4^{-2} + 3*4^{-3} = 0.984375$
0	0	0	1	.	0	0	0	1
...								
3	3	3	3	.	3	3	0	$3*4^3 + 3*4^2 + 3*4^1+3*4^0+3*4^{-1} + 3*4^{-2} + 0*4^{-3} = ?$
3	3	3	3	.	3	3	1	$3*4^3 + 3*4^2 + 3*4^1+3*4^0+3*4^{-1} + 3*4^{-2} + 1*4^{-3} = ?$
3	3	3	3	.	3	3	2	$3*4^3 + 3*4^2 + 3*4^1+3*4^0+3*4^{-1} + 3*4^{-2} + 2*4^{-3} = ?$
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

Solution:

A. More in m

B. Find the closest number

Radix-4								Radix-10
Integer (n=4)					Fraction (m=3)			
4^3	4^2	4^1	4^0	.	4^{-1}	4^{-2}	4^{-3}	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1*4^{-3} = 1/64 = 0.015625$
0	0	0	0	.	0	0	2	$2*4^{-3} = 2/64 = 0.03125$
0	0	0	0	.	0	0	3	$3*4^{-3} = 3/64 = 0.046875$
0	0	0	0	.	0	1	0	$1*4^{-2} + 0*4^{-2} = 1/16 = 0.0625$
0	0	0	0	.	0	1	1	$1*4^{-2} + 1*4^{-2} = 1/16 + 1/64 = 0.078125$
0	0	0	0	.	0	1	2	$1*4^{-2} + 2*4^{-2} = 1/16 + 2/64 = 0.09375$
0	0	0	0	.	0	1	3	$1*4^{-2} + 3*4^{-2} = 1/16 + 3/64 = 0.109375$
0	0	0	0	.	0	2	0	$2*4^{-2} + 0*4^{-2} = 2/16 = 0.125$
				...				
0	0	0	0	.	3	3	3	$3*4^{-1} + 3*4^{-2} + 3*4^{-3} = 0.984375$
0	0	0	1	.	0	0	0	1
0	0	0	1	.	0	0	1	1.015625
0	0	0	1	.	0	0	2	1.03125
				...				
3	3	3	3	.	3	3	2	$3*4^3 + 3*4^2 + 3*4^1+3*4^0+3*4^{-1} + 3*4^{-2} + 2*4^{-3} = ?$
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

Solution:

A. More in m

B. Find the closest number

$(1.001)_4 = (1.015625)_{10}$ => Error = 0.004375

$(1.002)_4 = (1.03125)_{10}$ => Error = 0.01125

Radix-4								Radix-10
Integer (n=4)					Fraction (m=3)			
4^3	4^2	4^1	4^0	.	4^{-1}	4^{-2}	4^{-3}	
0	0	0	0	.	0	0	0	0
0	0	0	0	.	0	0	1	$1*4^{-3} = 1/64 = 0.015625$
0	0	0	0	.	0	0	2	$2*4^{-3} = 2/64 = 0.03125$
0	0	0	0	.	0	0	3	$3*4^{-3} = 3/64 = 0.046875$
0	0	0	0	.	0	1	0	$1*4^{-2} + 0*4^{-2} = 1/16 = 0.0625$
0	0	0	0	.	0	1	1	$1*4^{-2} + 1*4^{-2} = 1/16 + 1/64 = 0.078125$
0	0	0	0	.	0	1	2	$1*4^{-2} + 2*4^{-2} = 1/16 + 2/64 = 0.09375$
0	0	0	0	.	0	1	3	$1*4^{-2} + 3*4^{-2} = 1/16 + 3/64 = 0.109375$
0	0	0	0	.	0	2	0	$2*4^{-2} + 0*4^{-2} = 2/16 = 0.125$
…								
0	0	0	0	.	3	3	3	$3*4^{-1} + 3*4^{-2} + 3*4^{-3} = 0.984375$
0	0	0	1	.	0	0	0	1
0	0	0	1	.	0	0	1	1.015625
0	0	0	1	.	0	0	2	1.03125
…								
3	3	3	3	.	3	3	2	$3*4^3 + 3*4^2 + 3*4^1+3*4^0+3*4^{-1} + 3*4^{-2} + 2*4^{-3} = ?$
3	3	3	3	.	3	3	3	255.984375

Is it possible to show the number $(1.02)_{10}$ in this system with these spaces?

No! Why?

Solution:

A. More in m

B. Find the closest number

$(1.001)_4 = (1.015625)_{10}$ => Error = 0.004375

$(1.002)_4 = (1.03125)_{10}$ => Error = 0.01125

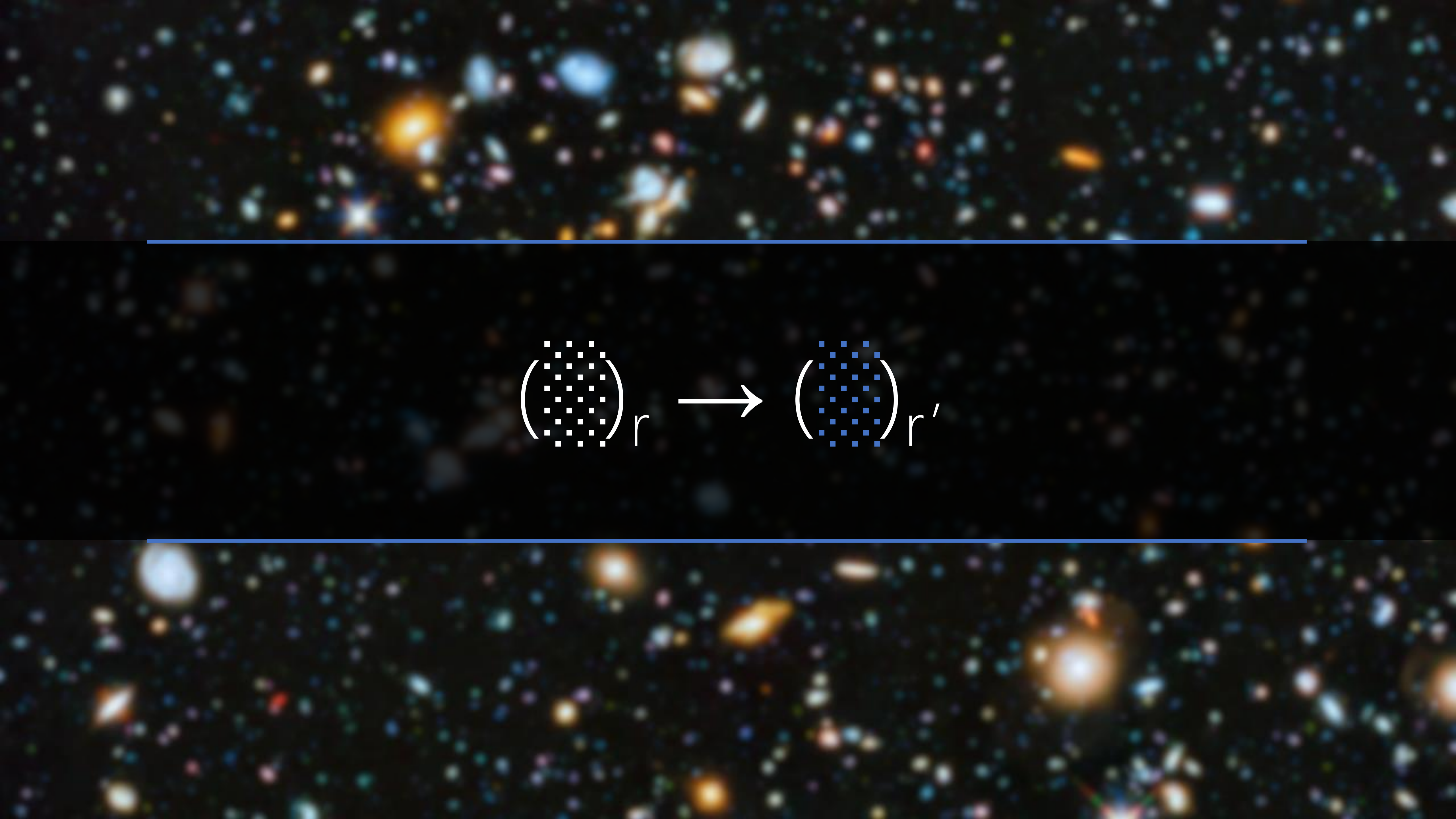
PRACTICE RADIX-[8,10,16]

At Home

INTEGER → INTEGER.FRACTION

CONVERSION

From Base-r to Base-r′


$(\;)_r \longrightarrow (\;)_{r'}$

$$(\cdots)_r \longrightarrow (\cdots)_{10}$$

we already knew that: sum of the powers of r

Let $(N)_r$ be a radix-r (base-r) number in a positional weighting number system, then

$$(N)_r = (d_{n-1}\, r^{n-1} + \cdots + d_0 r^0 \,.\, d_{-1}\, r^{-1} + d_{-2}\, r^{-2} + \cdots + d_{-m}\, r^{-m})_{10}$$

Fraction Point

where:

- r = radix (base)
- d_i = digit at position i, $0 \le d_i \le r - 1$
- r^i = weight of position i
- n = number of digits in integer part of N
- m = number of digits in fraction part of N

$$(\ldots)_r \leftarrow (\ldots)_{10}$$

Generating the digits of number in base-r

$$(\quad)_{10} \leftarrow (\quad)_{10}$$

To Come up with an Algorithm!

What is the digit in the position with significance of 10^0?

$(\quad 3 \quad 0 \quad 3 \quad 0 \quad 2 \quad 1 \quad 3 \quad \boxed{1} \quad)_{10}$

Divide by 10^1, keep the remainder!

What is the digit in the position with significance of 10^1?

$$(\quad 3 \quad 0 \quad 3 \quad 0 \quad 2 \quad 1 \quad 3 \quad 1 \quad)_{10}$$

Divide by 10^2?

What is the digit in the position with significance of 10^2?

$$(\quad 3 \quad 0 \quad 3 \quad 0 \quad 2 \quad \boxed{1} \quad 3 \quad 1 \quad)_{10}$$

Divide by 10^3?

What is the digit in the position with significance of 10^i?

$$(30302131)_{10} \rightarrow 10^1 \rightarrow 1$$
$$(30302131)_{10} \rightarrow 10^2 \rightarrow 3$$
$$(30302131)_{10} \rightarrow 10^3 \rightarrow 1$$
$$(30302131)_{10} \rightarrow 10^4 \rightarrow 2$$
$$(30302131)_{10} \rightarrow 10^5 \rightarrow 0$$
$$(30302131)_{10} \rightarrow 10^6 \rightarrow 3$$
$$(30302131)_{10} \rightarrow 10^7 \rightarrow 0$$
$$(30302131)_{10} \rightarrow 10^8 \rightarrow 3$$

Divide by 10?

$$\left(\overbrace{3 \quad 0 \quad 3 \quad 0 \quad 2 \quad 1 \quad 3 \quad 1} \right)_{10}$$

$$30{,}302{,}131 \div 10 = 3{,}030{,}213$$
$$30{,}302{,}131 \,\%\, 10 = 1$$

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \;\, r\, 1$$

Divide by 10?

$$(\overbrace{3\quad 0\quad 3\quad 0\quad 2\quad 1\quad 3}\quad 1)_{10}$$

$$3{,}030{,}213 \div 10 = 303{,}021$$
$$3{,}030{,}213 \;\%\; 10 = 3$$

$$\frac{3{,}030{,}213}{10} = 303{,}021 \;\; r\;3$$

$$\left(\overbrace{3\quad 0\quad 3\quad 0\quad 2\quad 1}^{\text{Divide by 10?}}\quad 3\quad 1\right)_{10}$$

$$303{,}021 \div 10 = 30{,}302$$
$$303{,}021 \;\%\; 10 = 1$$

$$\frac{303{,}021}{10} = 30{,}302\ r\ 1$$

What is the digit in the position with significance of 10^i?

$$(30302131)_{10} \rightarrow 10^1 \rightarrow 1$$
$$(3030213)_{10} \rightarrow 10^1 \rightarrow 3$$
$$(303021)_{10} \rightarrow 10^1 \rightarrow 1$$
$$(30302)_{10} \rightarrow 10^1 \rightarrow 2$$
$$(3030)_{10} \rightarrow 10^1 \rightarrow 0$$
$$(303)_{10} \rightarrow 10^1 \rightarrow 3$$
$$(30)_{10} \rightarrow 10^1 \rightarrow 0$$
$$(3)_{10} \rightarrow 10^1 \rightarrow 3$$

What is the digit in the position with significance of 10^0?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \ \ r\ 1$$

Remainder of the 0+1 division by 10!

What is the digit in the position with significance of 10^1?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \;\; r\ 1$$

$$\frac{\phantom{3{,}030{,}213}}{10} = 303{,}021 \;\; r\ 3$$

Remainder of the 1+1 division by 10!

What is the digit in the position with significance of 10^2?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \;\; r\ 1$$

$$\frac{3{,}030{,}213}{10} = 303{,}021 \;\; r\ 3$$

$$\frac{303{,}021}{10} = 30{,}302 \;\; r\ 1$$

Remainder of the 2+1 division by 10!

What is the digit in the position with significance of 10^3?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \;\; r\ 1$$

$$\frac{\phantom{3{,}030{,}213}}{10} = 303{,}021 \;\; r\ 3$$

$$\frac{\phantom{303{,}021}}{10} = 30{,}302 \;\; r\ 1$$

$$\frac{\phantom{30{,}302}}{10} = 3{,}030 \;\; r\ 2$$

Remainder of the 3+1 division by 10!

What is the digit in the position with significance of 10^4?

$$\cfrac{\cfrac{\cfrac{\cfrac{\cfrac{30{,}302{,}131}{10} = 3{,}030{,}213 \ \ r\ 1}{10} = 303{,}021 \ \ r\ 3}{10} = 30{,}302 \ \ r\ 1}{10} = 3{,}030 \ \ r\ 2}{10} = 303 \ \ r\ 0$$

Remainder of the 4+1 division by 10!

What is the digit in the position with significance of 10^5?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \;\; r\ 1$$
$$\frac{3{,}030{,}213}{10} = 303{,}021 \;\; r\ 3$$
$$\frac{303{,}021}{10} = 30{,}302 \;\; r\ 1$$
$$\frac{30{,}302}{10} = 3{,}030 \;\; r\ 2$$
$$\frac{3{,}030}{10} = 303 \;\; r\ 0$$
$$\frac{303}{10} = 30 \;\; r\ 3$$

Remainder of the 5+1 division by 10!

What is the digit in the position with significance of 10^6?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \;\; r\, 1$$
$$\frac{3{,}030{,}213}{10} = 303{,}021 \;\; r\, 3$$
$$\frac{303{,}021}{10} = 30{,}302 \;\; r\, 1$$
$$\frac{30{,}302}{10} = 3{,}030 \;\; r\, 2$$
$$\frac{3{,}030}{10} = 303 \;\; r\, 0$$
$$\frac{303}{10} = 30 \;\; r\, 3$$
$$\frac{30}{10} = 3 \;\; r\, 0$$

Remainder of the 6+1 division by 10!

What is the digit in the position with significance of 10^7?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \ r\ 1$$
$$\frac{3{,}030{,}213}{10} = 303{,}021 \ r\ 3$$
$$\frac{303{,}021}{10} = 30{,}302 \ r\ 1$$
$$\frac{30{,}302}{10} = 3{,}030 \ r\ 2$$
$$\frac{3{,}030}{10} = 303 \ r\ 0$$
$$\frac{303}{10} = 30 \ r\ 3$$
$$\frac{30}{10} = 3 \ r\ 0$$
$$\frac{3}{10} = 0 \ r\ 3$$

Remainder of the 7+1 division by 10!

What is the digit in the position with significance of 10^i?

$$\frac{30{,}302{,}131}{10} = 3{,}030{,}213 \; r \; 1$$

$$\frac{3{,}030{,}213}{10} = 303{,}021 \; r \; 3$$

$$\frac{303{,}021}{10} = 30{,}302 \; r \; 1$$

$$\frac{30{,}302}{10} = 3{,}030 \; r \; 2$$

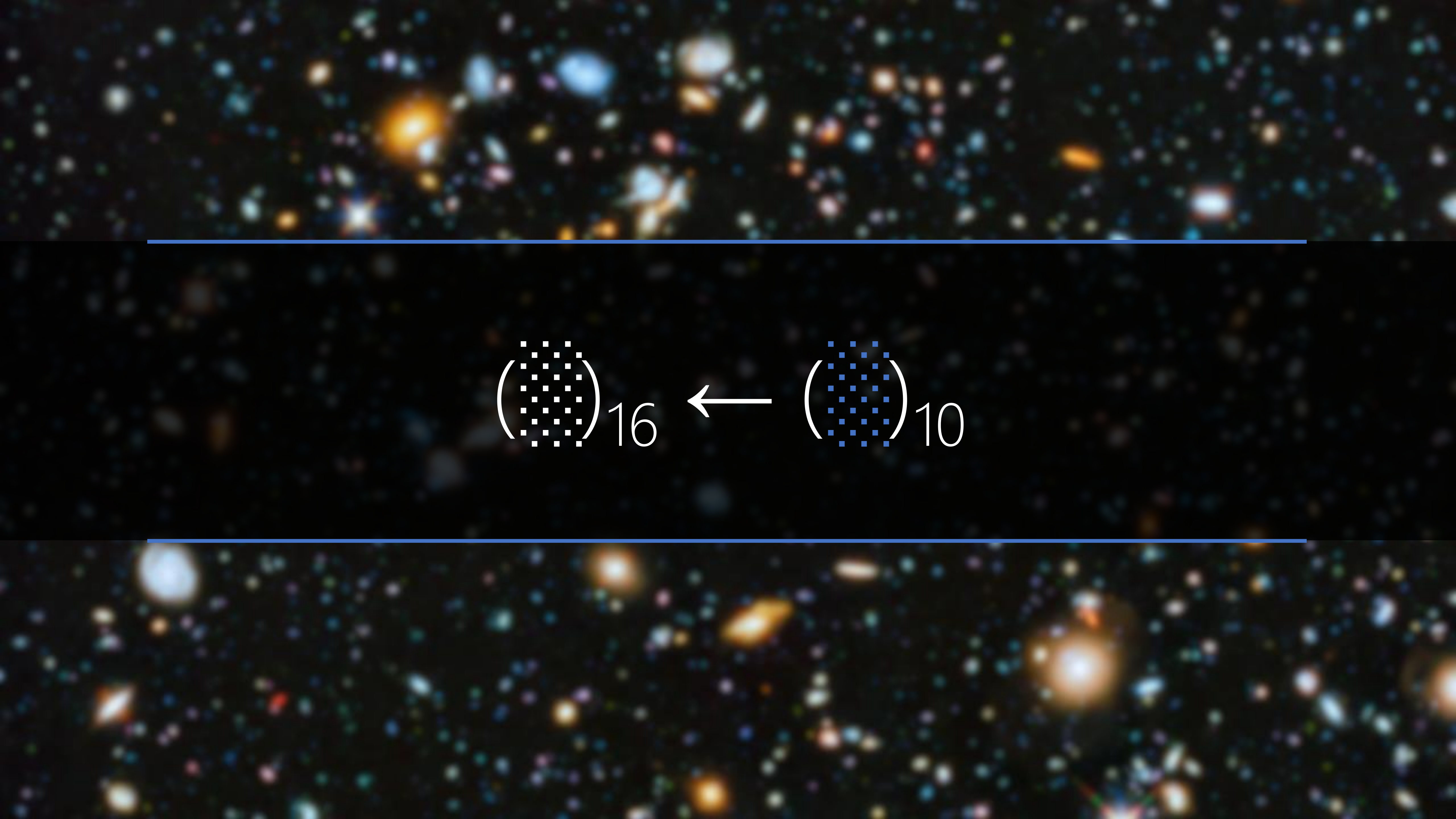
$$\frac{3{,}030}{10} = 303 \; r \; 0$$

$$\frac{303}{10} = 30 \; r \; 3$$

$$\frac{30}{10} = 3 \; r \; 0$$

$$\frac{3}{10} = 0 \; r \; 3$$

$$(\;3\quad 0\quad 3\quad 0\quad 2\quad 1\quad 3\quad 1\;)_{10}$$



$$(\quad)_{16} \leftarrow (\quad)_{10}$$

What is the digit in the position with significance of 16^0?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3$$

Remainder of the 0+1 division by 16!

What is the digit in the position with significance of 16^1?

$$\cfrac{\cfrac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3}{16} = 118{,}367\ r\ 11$$

Remainder of the 1+1 division by 16!

What is the digit in the position with significance of 16^2?

$$\cfrac{\cfrac{\cfrac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3}{16} = 118{,}367\ r\ 11}{16} = 7{,}397\ r\ 15$$

Remainder of the 2+1 division by 16!

What is the digit in the position with significance of 16^3?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3$$

$$\frac{\qquad}{16} = 118{,}367\ r\ 11$$

$$\frac{\qquad}{16} = 7{,}397\ r\ 15$$

$$\frac{\qquad}{16} = 462\ r\ 5$$

Remainder of the 3+1 division by 16!

What is the digit in the position with significance of 16^4?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3$$

$$\frac{\quad}{16} = 118{,}367\ r\ 11$$

$$\frac{\quad}{16} = 7{,}397\ r\ 15$$

$$\frac{\quad}{16} = 462\ \ r\ 5$$

$$\frac{\quad}{16} = 28\ \ r\ 14$$

Remainder of the 4+1 division by 16!

What is the digit in the position with significance of 16^5?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3$$

$$\frac{}{16} = 118{,}367\ r\ 11$$

$$\frac{}{16} = 7{,}397\ r\ 15$$

$$\frac{}{16} = 462\ \ r\ 5$$

$$\frac{}{16} = 28\ \ r\ 14$$

$$\frac{}{16} = 1\ \ r\ 12$$

Remainder of the 5+1 division by 16!

What is the digit in the position with significance of 16^6?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3$$

$$\frac{\phantom{1{,}893{,}883}}{16} = 118{,}367\ r\ 11$$

$$\frac{\phantom{118{,}367}}{16} = 7{,}397\ r\ 15$$

$$\frac{\phantom{7{,}397}}{16} = 462\ r\ 5$$

$$\frac{}{16} = 28\ r\ 14$$

$$\frac{}{16} = 1\ r\ 12$$

$$\frac{}{16} = 0\ r\ 1$$

Remainder of the 6+1 division by 16!

What is the digit in the position with significance of 16^6?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 3$$

$$\frac{\qquad}{16} = 118{,}367\ r\ 11$$

$$\frac{\qquad}{16} = 7{,}397\ r\ 15$$

$$\frac{\qquad}{16} = 462\ \ r\ 5$$

$$\frac{\qquad}{16} = 28\ \ r\ 14$$

$$\frac{\qquad}{16} = 1\ \ r\ 12$$

$$\frac{\qquad}{16} = 0\ \ r\ 1$$

$$(\quad 1 \quad 12 \quad 14 \quad 5 \quad 15 \quad 11 \quad 3 \quad)_{16}$$

What is the digit in the position with significance of 16^6?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883\ r\ 1$$

$$\frac{}{16} = 118{,}367\ r\ 11$$

$$\frac{}{16} = 7{,}397\ r\ 0$$

$$\frac{}{16} = 462\ r\ 5$$

$$\frac{}{16} = 28\ r\ 14$$

$$\frac{}{16} = 1\ r\ 12$$

$$\frac{}{16} = 0\ r\ 1$$

$$(\ 1\quad 12\quad 14\quad 5\quad 15\quad 11\quad 3\ )_{16}$$

$$1,2,3,4,5,6,7,8,9,A = 9 + 1 = (10)_{10}$$

$$B = A + 1 = (11)_{10}$$

$$C = B + 1 = (12)_{10}$$

$$D = C + 1 = (13)_{10}$$

$$E = D + 1 = (14)_{10}$$

$$F = E + 1 = (15)_{10}$$

What is the digit in the position with significance of 16^6?

$$\frac{30{,}302{,}131}{16} = 1{,}893{,}883 \; r \; 1$$

$$\frac{\phantom{1{,}893{,}883}}{16} = 118{,}367 \; r \; 11$$

$$\frac{\phantom{118{,}367}}{16} = 7{,}397 \; r \; 0$$

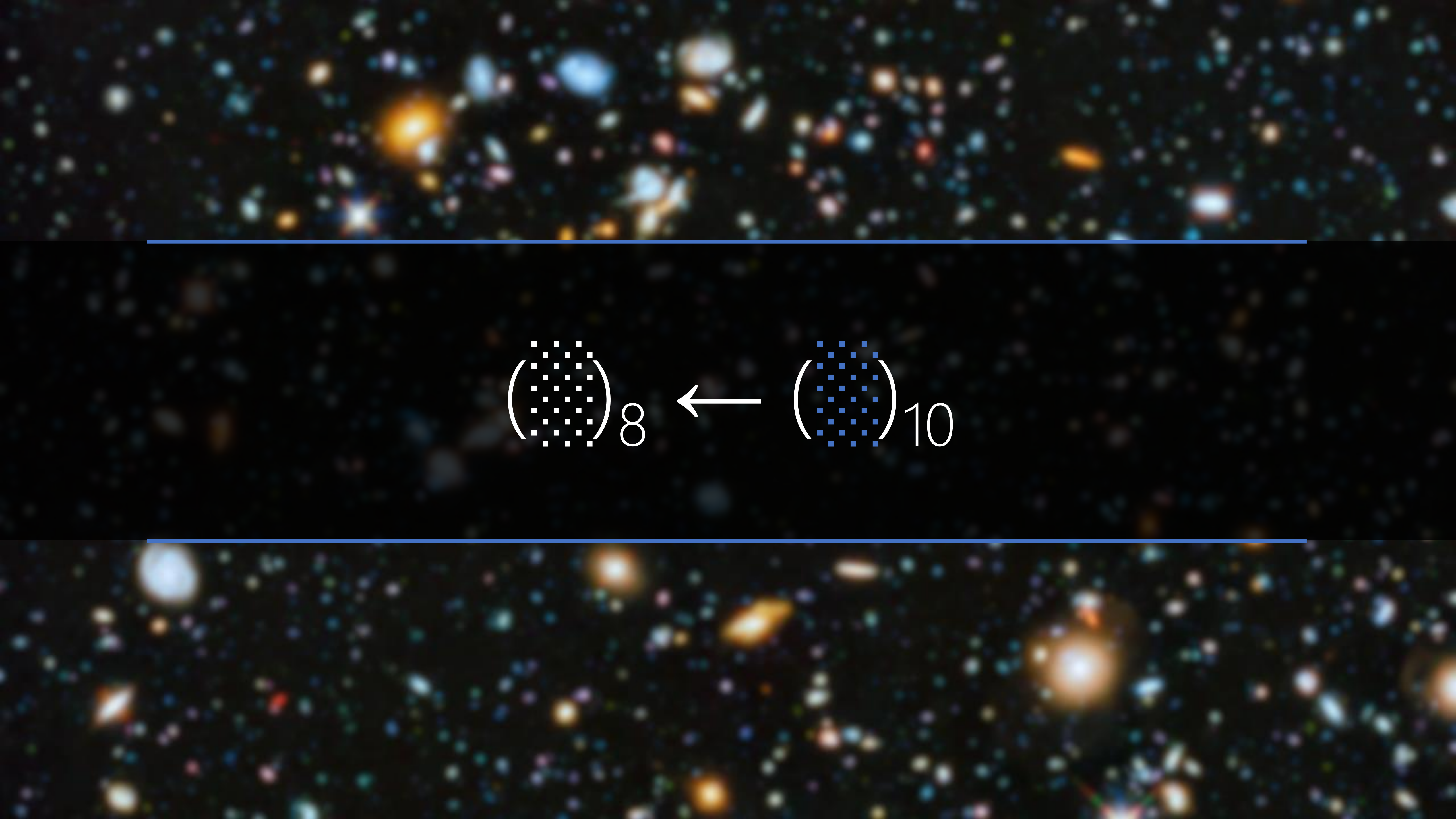
$$\frac{\phantom{7{,}397}}{16} = 462 \; r \; 5$$

$$\frac{}{16} = 28 \; r \; 14$$

$$\frac{}{16} = 1 \; r \; 12$$

$$\frac{}{16} = 0 \; r \; 1$$

$(\;1\;\;C\;\;E\;\;5\;\;F\;\;B\;\;3\;)_{16}$


8
←
10

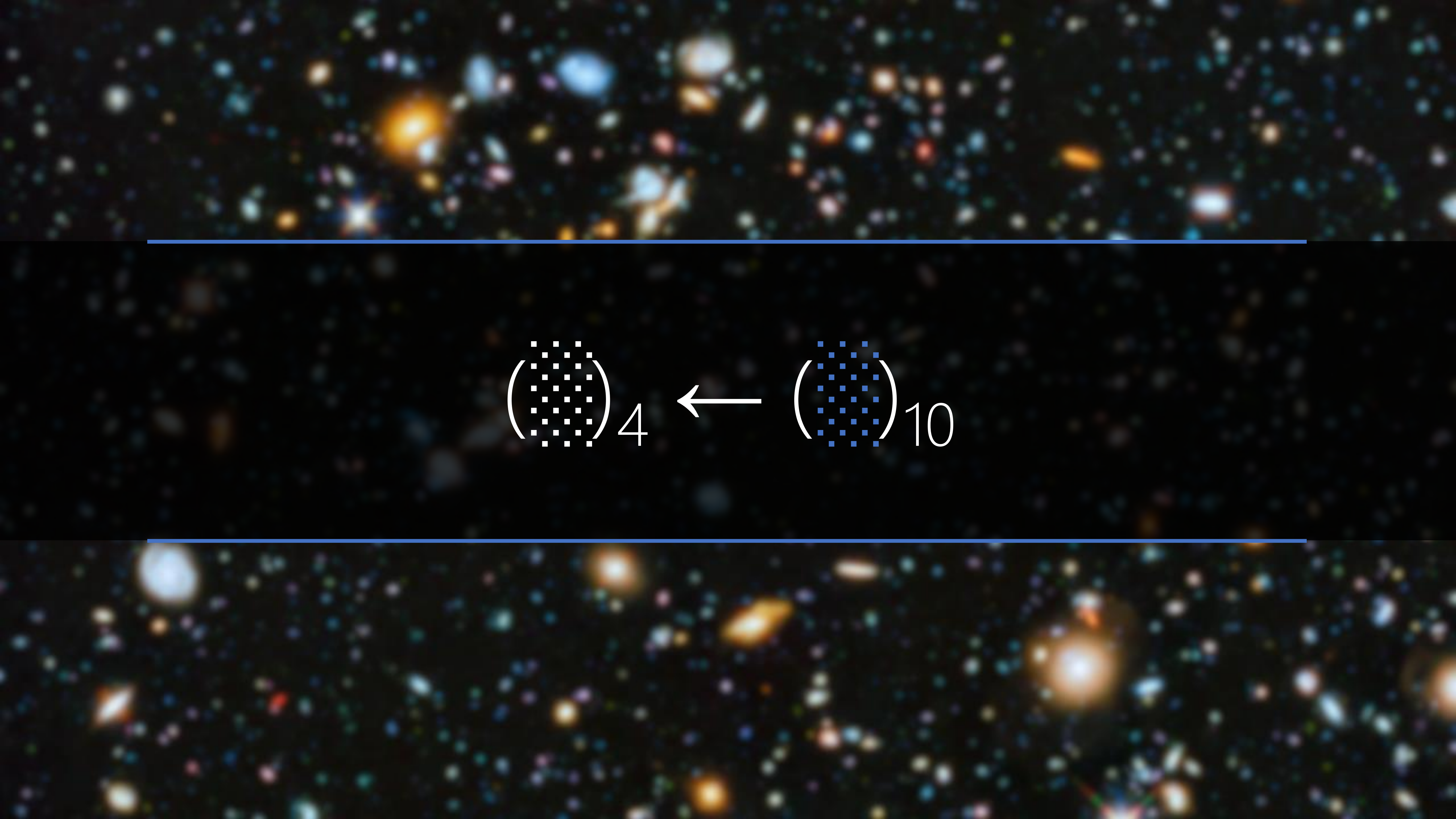
What is the digit in the position with significance of 8^i?

Quotient	Remainder
30,302,131÷8	3
3787766÷8	6
473470÷8	6
59183÷8	7
7397÷8	5
924÷8	4
115÷8	3
14÷8	6
1÷8	1
0	

$(1CE5FB3)_{16}$

$(30{,}302{,}131)_{10}$

$(163457663)_8$


$(\quad)_{4} \leftarrow (\quad)_{10}$

What is the digit in the position with significance of 4^i?

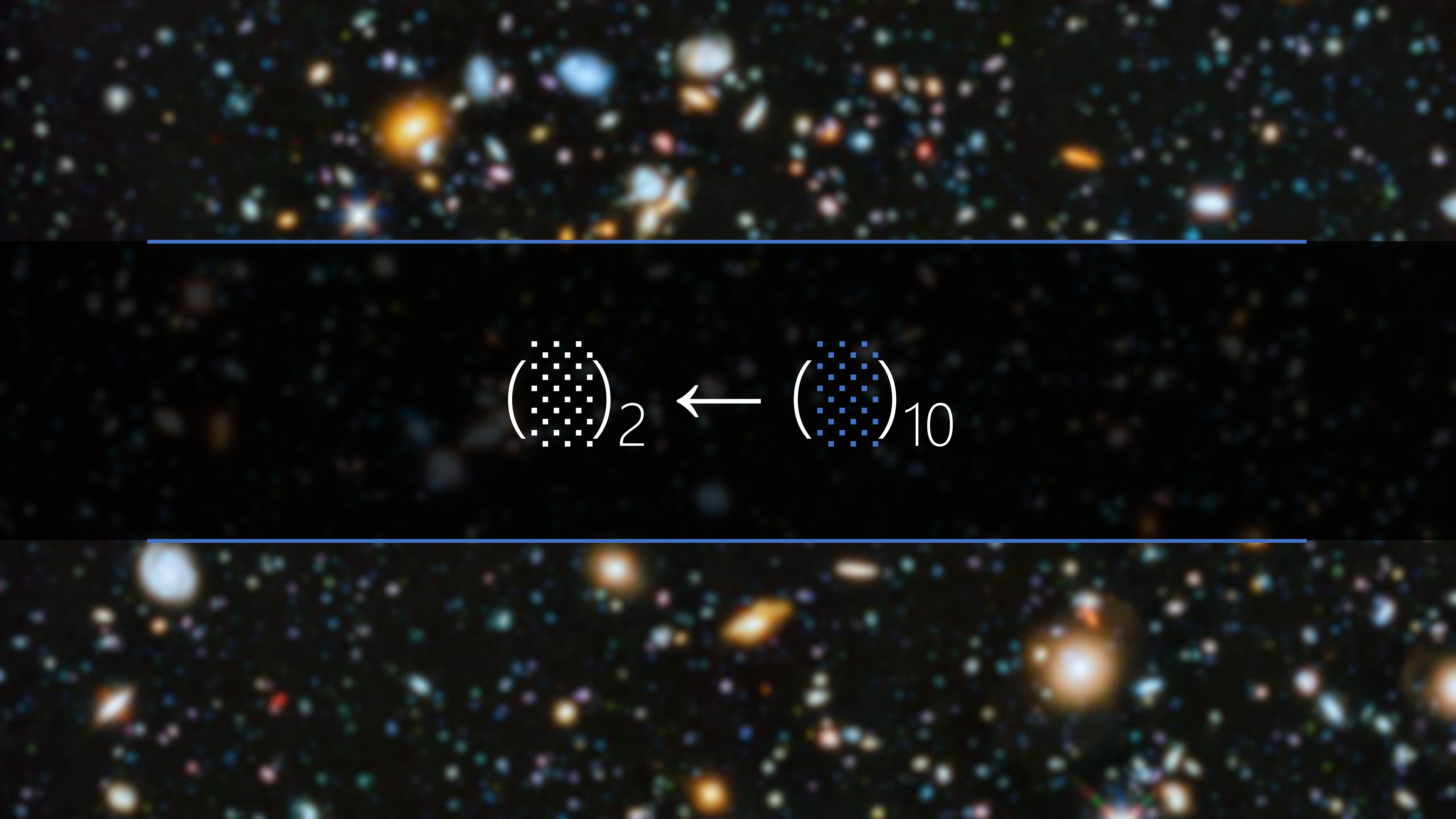
Quotient	Remainder
30,302,131÷4	3
7575532÷4	0
1893883÷4	3
473470÷4	2
118367÷4	3
29591÷4	3
7397÷4	1
1849÷4	1
462÷4	2
115÷4	3
28÷4	0
7÷4	3
1÷4	1
0	

$(1CE5FB3)_{16}$

$(30,302,131)_{10}$

$(163457663)_8$

$(1303211332303)_4$


$(\quad)_2 \leftarrow (\quad)_{10}$

What is the digit in the position with significance of 2^i?

Quotient	Remainder
30,302,131÷2	1
15151065÷2	1
7575532÷2	0
3787766÷2	0
1893883÷2	1
946941÷2	1
473470÷2	0
236735÷2	1
118367÷2	1
59183÷2	1
29591÷2	1
14795÷2	1
7397÷2	1
3698÷2	0
1849÷2	1
924÷2	0
462÷2	0
231÷2	1
115÷2	1
57÷2	1
28÷2	0
14÷2	0
7÷2	1
3÷2	1
1÷2	1
0	

$(1CE5FB3)_{16}$

$(30{,}302{,}131)_{10}$

$(163457663)_{8}$

$(1303211332303)_{4}$

$(1110011100101111110110011)_{2}$

What is the digit in the position with significance of 2^i?

Quotient	Remainder
30,302,131÷2	1
15151065÷2	1
7575532÷2	0
3787766÷2	0
1893883÷2	1
946941÷2	1
473470÷2	0
236735÷2	1
118367÷2	1
59183÷2	1
29591÷2	1
14795÷2	1
7397÷2	1
3698÷2	0
1849÷2	1
924÷2	0
462÷2	0
231÷2	1
115÷2	1
57÷2	1
28÷2	0
14÷2	0
7÷2	1
3÷2	1
1÷2	1
0	

$(1CE5FB3)_{16}$

$(30,302,131)_{10}$

$(163457663)_8$

$(1303211332303)_4$

$(1110011100101111110110011)_2$

Less space the better!

But why binary took the stage?

$$(\cdots)_r \leftarrow (\cdots)_{10}$$

1. Divide quotients by r
2. Keep the remainders as the digits in r

For r > 10, substitute the digits with valid digits, e.g., in Base-16: 15→F

$$(\cdots)_r \longrightarrow (\cdots)_{r'}$$

Any Base-r to Any Base-r′

Mediate through Base-10

Base-r ⟶ Base-10 ⟶ Base-r′

$$(1CE5FB3)_{16} \rightarrow 1\times16^6+C\times16^5+E\times16^4+5\times16^3+F\times16^2+B\times16^1+3\times16^0$$

$$\rightarrow 1\times16^6+12\times16^5+14\times16^4+5\times16^3+15\times16^2+11\times16^1+3\times16^0$$

$$\rightarrow (30{,}302{,}131)_{10}$$

Quotient	Remainder
30,302,131÷8	3
3787766÷8	6
473470÷8	6
59183÷8	7
7397÷8	5
924÷8	4
115÷8	3
14÷8	6
1÷8	1
0	

$$\rightarrow (163457663)_8$$

FRACTION CONVERSION

$$(0.\cdots)_{10} \leftarrow (0.\cdots)_{10}$$

To Come up with an Algorithm!

What is the digit in the position with significance of 10^{-i}?

$$(0.26501)_{10} \rightarrow 10^{-1} \rightarrow 2$$
$$(0.26501)_{10} \rightarrow 10^{-2} \rightarrow 6$$
$$(0.26501)_{10} \rightarrow 10^{-3} \rightarrow 5$$
$$(0.26501)_{10} \rightarrow 10^{-4} \rightarrow 0$$
$$(0.26501)_{10} \rightarrow 10^{-5} \rightarrow 1$$

What is the digit in the position with significance of 10^{-i}?

$$(0.26501)_{10} \rightarrow 10^{-1} \rightarrow 2$$

$$(0.6501)_{10} \rightarrow 10^{-1} \rightarrow 6$$

$$(0.501)_{10} \rightarrow 10^{-1} \rightarrow 5$$

$$(0.01)_{10} \rightarrow 10^{-1} \rightarrow 0$$

$$(0.1)_{10} \rightarrow 10^{-1} \rightarrow 1$$

Divide by 10^{-1} = Multiply by 10?

$$(\underbrace{\quad 0 \quad . \quad}\overbrace{2 \quad 6 \quad 5 \quad 0 \quad 1})_{10}$$

$$0.26501 \div 10^{-1} = 0.26501 \times 10 = 2.6501$$

Divide by 10^{-1} = Multiply by 10?

$$(0\;.\;2\;6\;5\;0\;1)_{10}$$

$$0.6501 \div 10^{-1} = 0.6501 \times 10 = 6.501$$

Fraction	Result	Integer Part
0.26501×10	2.6501	2
0.6501×10	6.501	6
0.501×10	5.01	5
0.01×10	0.1	0
0.1×10	1.0	1
0		

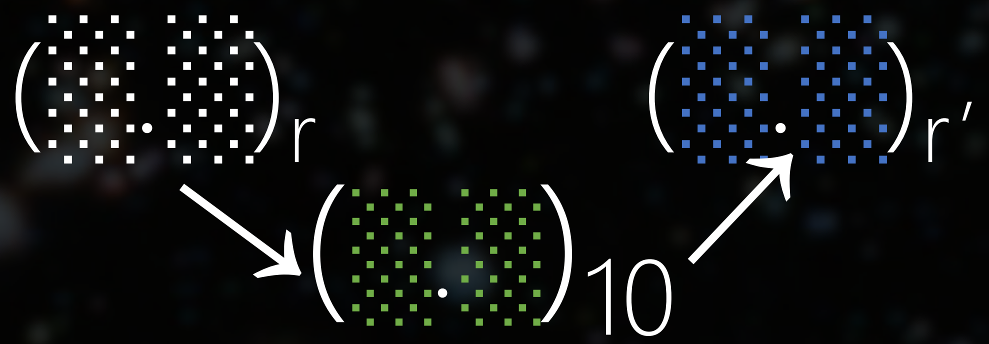
$$(0.\,)_{16} \leftarrow (0.\,)_{10}$$

Fraction	Result	Integer Part
0.26501×16	4.24016	4
0.24016×16	3.84256	3
0.84256×16	13.48096	13 = D
0.48096×16	7.69536	7
0.69536×16	11.12576	11 = B
0.12576×16	2.01216	2
0.01216×16	0.19456	0
0.194560006×16	3.11296	3
0.1129601×16	1.807362	1
0.807361603×16	12.91779	12 = C
0.917785645×16	14.68457	14 = E
0.684570313×16	10.95313	10 = A
0.953125×16	15.25	15 = F
0.25×16	4	4
0	0	0

$$(0.26501)_{10} \rightarrow (0.43D7B2031CEAF40)_{16}$$

$$(0.\ldots)_8 \leftarrow (0.\ldots)_{10}$$
$$(0.\ldots)_4 \leftarrow (0.\ldots)_{10}$$
$$(0.\ldots)_2 \leftarrow (0.\ldots)_{10}$$

MICROSOFT EXCEL SHEET



Integer Part independent of Fraction Part!

$$(\;\;.)_r \rightarrow (\;\;.)_{10} \rightarrow (\;\;.)_{r'}$$

Integer Part

$$(.\ \square)_r \rightarrow (.\ \square)_{10} \rightarrow (.\ \square)_{r'}$$

Fraction Part

$(30,302,131.\ 26501)_{10}$ $(163457663.20753....)_8$

Quotient	Remainder
30,302,131÷8	3
3787766÷8	6
473470÷8	6
59183÷8	7
7397÷8	5
924÷8	4
115÷8	3
14÷8	6
1÷8	1
0	

Fraction	Result	Integer Part
0.26501×8	2.12008	2
0.12008×8	0.96064	0
0.96064×8	7.68512	7
0.68512×8	5.48096	5
0.48096×8	3.84768	3
0.84768×8	6.78144	6
0.78144×8	6.25152	6
0.25152×8	2.01216	2
0.01216×8	0.09728	0
…	…	…

BASE-2 → BASE-[2^2, 2^3, 2^4, 2^n]

LECTURE ASSIGNMENT

BASE-2 ← BASE-[2^2, 2^3, 2^4, 2^n]

LECTURE ASSIGNMENT

https://planetcalc.com/862/